

Designed by Remus Teodorescu 2026

LECTURE 9.1

Laminar Flow

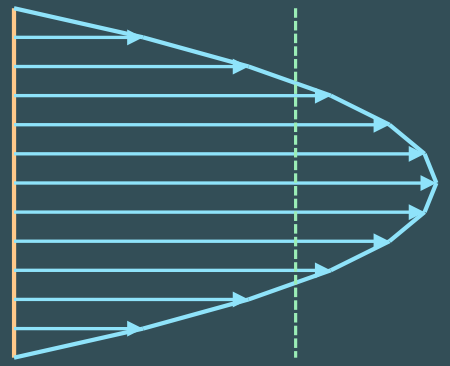
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Deep Learning for Engineering · 2026

NEWTON · THE PIPE · DARCY



Navier–Stokes is Newton's law for a parcel of liquid. In a straight tube far from its bends it reduces to one equation, solved by hand: Hagen–Poiseuille's parabola, Darcy's law, and the start-up when the pump begins. FDM and a PINN then solve it against the truth, and a flow meter gives the viscosity.

Learning Objectives

SECTIONS, IN ORDER

1. **Flow equations** — Newton's law for a liquid, the Reynolds number
2. **Pipe flow** — fully developed, Hagen–Poiseuille and Darcy, the start-up
3. **Solving a flow** — conditions built in, FDM or PINN, the viscosity

White carries the pipe flow and Liu the method; the example is the exercise's own, the coolant in a 6 mm tube of a cold plate.

READ, AND WHAT EACH ANSWERS

White ch. 4	what does Newton say for a liquid?
White ch. 6	what is the flow in a tube?
Liu ch. 2.2	how is it solved, and checked?

Recap of L8.2 Transient Heat

1 – 2 · A NETWORK IN TIME

1 **How does a network represent the temperature of the heat sink in time, and how is its initial temperature made exact?**

Time becomes one more input, and the whole block of space and time is trained at once rather than marched, so the temperature can be read at any instant without running again. As a loss term the initial temperature is one slice among many interior points and is easily outvoted. In Ex_08 it is built into the trial function by a factor of the scaled time that is zero at the start, so the initial field holds for any weights. *Liu ch. 4.2 – 4.3; Ex_08 section 11*

2 **Why does the loss of a network in time not respect causality, and how is training steered towards the early instants?**

The loss is one mean over the block of space and time and treats an early instant exactly as a late one, so a network can fit the late times first and carry an error at the start, the one instant known before training. The remedies, in order: build the start in, weight each instant by how well its past is fitted, train window after window, or sample early times densely. Ex_08 builds the start in and puts half its points in the first fifth of the window. *Liu ch. 4.4; Ex_08 section 11*

3 – 4 · THE PULSE AND THE INVERSE

3 **What does the network of the pulse have built in, and how does it compare with Crank–Nicolson?**

The trial function carries the lumped model of the whole sink warming and cooling, a factor of time that makes the start exact, and the cosine input that keeps the ends insulated; the network learns only the spreading. Its clock is the square root of the scaled time, which stretches the first seconds. Crank–Nicolson was within a few hundredths of a kelvin at every instant, the network one to two kelvin off during the pulse: the grid wins the forward transient. *Liu ch. 4.2 – 4.4; Ex_08 sections 9 – 11*

4 **How can the heat transfer coefficient of the fins be estimated with a network from one thermocouple, and what identifies it?**

The coefficient becomes one trainable number beside the weights, written as an exponential so that it stays positive; it enters the convection, the lumped model and the energy balance, and a data term matches the thermocouple readings. The rise during the pulse is set by the known heat capacity; the cooling after switch-off is set by the coefficient alone and identifies it. It was found 4.66 % below the true value. *Liu ch. 4.6; Ex_08 section 12*

Navier–Stokes

Navier–Stokes — Newton's second law for a small parcel of liquid: mass times acceleration equals the forces on it. The mass per volume is the density ρ; the acceleration is how the parcel's velocity changes as it moves; the forces are the pressure, the friction of the neighbours, the weight.

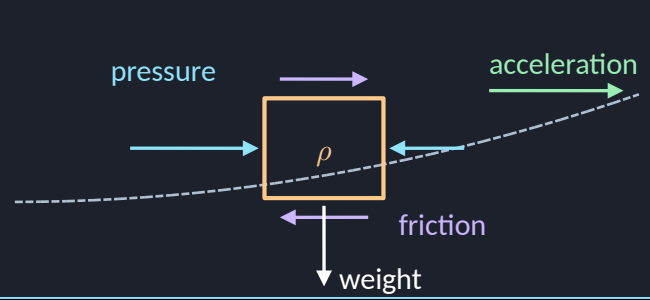
MASS TIMES ACCELERATION EQUALS THE FORCES

$$\rho \quad (\mathbf{u}_t + (\mathbf{u}\cdot\nabla)\mathbf{u}) \quad = \quad -\nabla p + \mu\nabla^2\mathbf{u} + \rho\,\mathbf{g}, \qquad \nabla\cdot\mathbf{u} = 0$$

mass — acceleration — the forces

ρ the mass per volume, $\mathbf{u}_t + (\mathbf{u}\cdot\nabla)\mathbf{u}$ the acceleration of the parcel : in time, and because it moves on
$-\nabla p$ the push of the pressure, $\mu\nabla^2\mathbf{u}$ the friction of the neighbours ($\mu = \rho\nu$), $\rho\mathbf{g}$ the weight, $\nabla\cdot\mathbf{u} = 0$ the volume kept

A PARCEL OF LIQUID, AND NEWTON'S LAW ON IT



mass: the density, per volume

acceleration: in time, and as it moves

forces: pressure, friction, weight

NEWTON'S LAW, TERM BY TERM

Newton	for a parcel of liquid	the term
mass	the density, per volume	ρ
acceleration	the velocity changes in time	$\mathbf{u}_t$
	and because it moves on	$(\mathbf{u}\cdot\nabla)\mathbf{u}$
force	the pressure, from the sides	$-\nabla p$
	the neighbours drag on it	$\mu\nabla^2\mathbf{u}$
	its weight	$\rho\mathbf{g}$

Why $(\mathbf{u}\cdot\nabla)\mathbf{u}$ is an acceleration: in a time dt the parcel moves from x to $x + \mathbf{u}\cdot dt$, where the velocity is different, so its velocity changes by $\mathbf{u}_t\cdot dt + (\mathbf{u}\cdot dt\cdot\nabla)\mathbf{u}$.

In a steady flow through a narrowing pipe $\mathbf{u}_t = 0$, yet every parcel speeds up.

Inertia and Darcy

Convective inertia term — $\rho(\mathbf{u}\cdot\nabla)\mathbf{u}$: momentum carried by the flow itself, so $\mathbf{u}$ appears twice. Send the same flow twice as fast: $\rho\mathbf{u}_t$ and $\mu\nabla^2\mathbf{u}$ hold $\mathbf{u}$ once and double, $\rho(2\mathbf{u}\cdot\nabla)(2\mathbf{u}) = 4\rho(\mathbf{u}\cdot\nabla)\mathbf{u}$ quadruples. So two flows do not add, and large eddies feed smaller ones: turbulence.

FRAME IT, NEGLECT IT, AVERAGE IT

$$\rho\,\mathbf{u}_t + \boxed{\rho\,(\mathbf{u}\cdot\nabla)\mathbf{u}} = -\nabla p + \mu\,\nabla^2\mathbf{u} + \rho\,\mathbf{g}$$

keep it: nonlinear, turbulence
$\rho\mathbf{u}_t$ stays: the local acceleration

$$\rho\,\mathbf{u}_t = -\nabla p + \mu\,\nabla^2\mathbf{u} + \rho\,\mathbf{g}$$

neglect it: Stokes, linear, slow flow

$$\bar{u} = -\frac{K}{\mu}\frac{dp}{dx}, \qquad K = \frac{h^2}{12}$$

average across the channel: Darcy
the law this lecture rests on

$(\mathbf{u}\cdot\nabla)\mathbf{u}$ the inertia, the one nonlinear term; $\bar{u}$ the mean speed across the channel, dp/dx the pressure drop per length
K the permeability, $h^2/12$ for a channel of height h, μ the viscosity, $\rho\mathbf{g}$ the weight

WHY IT IS NONLINEAR: SEND THE FLOW TWICE AS FAST

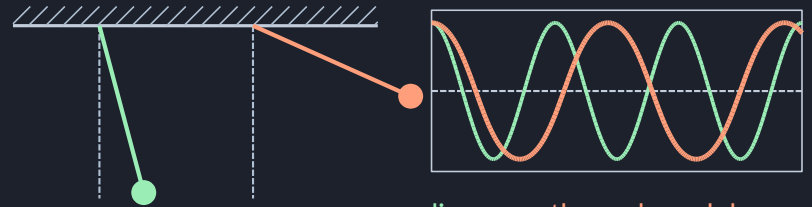
$\mathbf{u}$ once: $\rho\mathbf{u}_t$, $\mu\nabla^2\mathbf{u}$ double
$\rho\mathbf{g}$ unchanged, ∇p follows

$\mathbf{u}$ twice: $\rho(2\mathbf{u}\cdot\nabla)(2\mathbf{u}) = 4\rho(\mathbf{u}\cdot\nabla)\mathbf{u}$

THE SAME STEP ON A PENDULUM

$$\ddot{\theta} + \frac{g}{\ell}\sin\theta = 0 \quad \Rightarrow \quad \ddot{\theta} + \frac{g}{\ell}\theta = 0$$

θ the angle, ℓ the length, g gravity; $\sin\theta \to \theta$ is the same step as dropping the inertia



Small swing, sin $\theta \approx \theta$: linear, the same period at any amplitude - that is Stokes. Large swing: the period grows with the amplitude, driven it goes chaotic - that is Navier–Stokes. The amplitude decides whether the nonlinearity matters, as Re does for a flow.

The Navier–Stokes Millennium Problem

Millennium problem — given a smooth initial velocity field of finite energy, do the three-dimensional incompressible equations keep a smooth solution for all time, or can the speed become unbounded in finite time? One of the seven Clay Millennium Prize problems, stated so that a smooth applied force is allowed.

EULER FIRST, THEN NAVIER–STOKES

$$\mathbf{u}_t + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \mathbf{f} \qquad (\nu = 0)$$

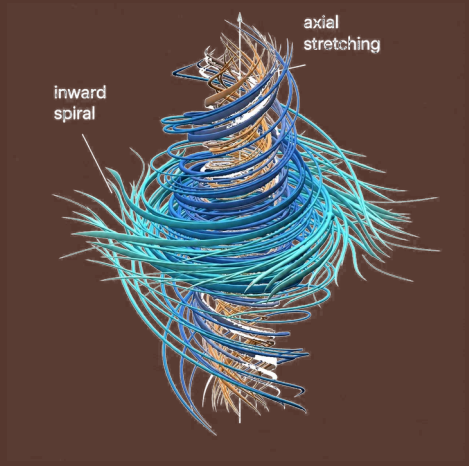
$$\mathbf{u}_t + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu\,\nabla^2\mathbf{u} + \mathbf{f}$$

u velocity, p pressure, ν viscosity, **f** an applied force; Euler is Navier-Stokes with no friction

THE ORDER OF THE MATHEMATICS, AND WHAT IS OPEN

- **Euler** — no friction: the easier case, broken first
- **Navier–Stokes with a force** — friction lost, machine-checked
- **open** — the unforced question; prize unclaimed

THE FLOW, AND WHO PROVED WHAT



the fluid spirals inward and is stretched along the axis; a thinner, faster core at every step, so the speed can grow without bound

who	what they proved
Martínez-Zoroa, Córdoba	the cascade, by hand
Alpöge, Buckmaster	forced Euler, announced first
OpenAI's system	unforced Euler, then forced N–S
Clay Institute	has not accepted it

The Reynolds Number

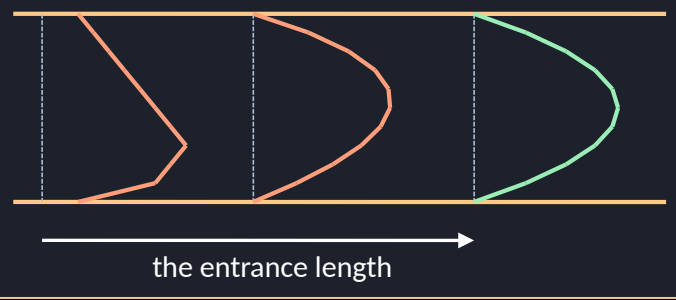
Reynolds number — the ratio of inertia to viscosity, $Re = \rho \bar{u} D / \mu$. In a pipe below about 2300 the flow stays laminar, layer sliding over layer; above it, it turns turbulent. It also sets how far the flow needs after a bend to settle into its profile: the entrance length, $0.05\ Re\ D$.

ONE NUMBER, AND THE LENGTH IT SETS

$$Re = \frac{\rho\, \bar{u}\, D}{\mu} = 452 \qquad L_e \approx 0.05\, Re\, D = 136 \text{ mm}$$

ρ the density, $\bar{u}$ the mean speed, D the diameter, μ the viscosity
L_e the entrance length : how far after a bend the profile needs to settle

AFTER A BEND, THE PROFILE SETTLES

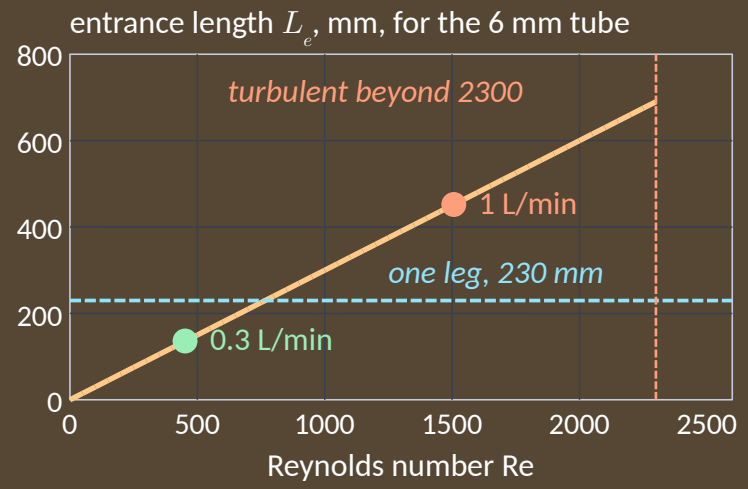


after a bend: skewed

settling along the straight leg

the parabola, fully developed

ENTRANCE LENGTH AGAINST Re



A leg is one straight run of the serpentine between two bends, 230 mm here. At 0.3 L/min the parabola forms 136 mm after each bend, inside the leg; at 1 L/min it would need 452 mm and never form.

Flow Boundary Conditions

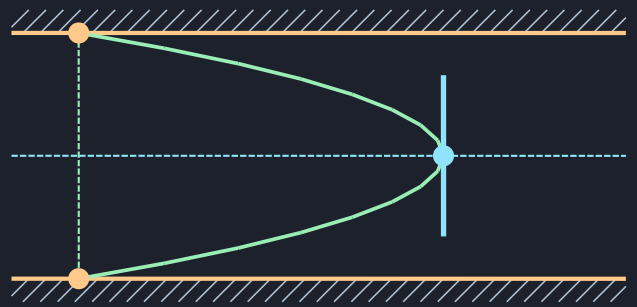
Flow boundary conditions — no-slip at a wall at rest: the fluid touching it is at rest too, $\mathbf{u} = 0$ - an assumption, confirmed by experiment. In a tube $u = 0$ at $r = R$; on the axis the profile is symmetric, so $u_r = 0$; at the start the coolant is at rest; the pump sets the pressure gradient $G = -\partial p/\partial x$.

WALL, AXIS, START, PUMP

$$u(R, t) = 0, \quad u_r(0, t) = 0, \quad u(r, 0) = 0$$

u the velocity along the tube, r the distance from the axis, R the radius
u_r its slope across : zero on the axis, where the profile is symmetric

A VALUE AT THE WALL, A SLOPE ON THE AXIS

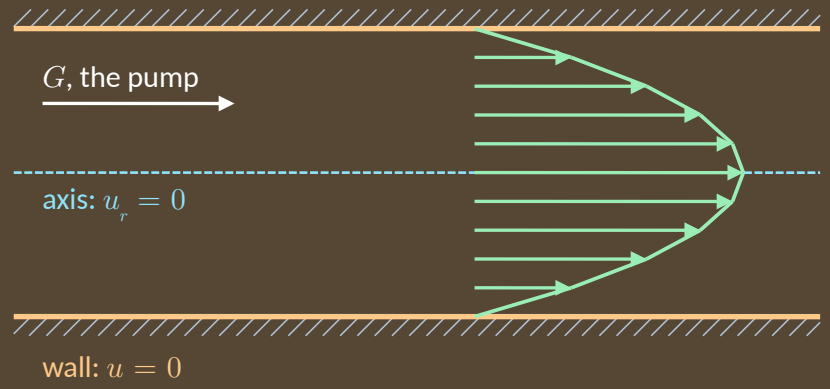


wall: value given (Dirichlet) - easy, built into the trial function

axis: slope given (Neumann) - harder: a loss term, or symmetry built in

so the parabola crosses the axis flat

THE TUBE OF EX_09.1, ALONG ITS AXIS



wall	$u = 0$: no slip, stuck not slowed
axis	$u_r = 0$: symmetry
start	$u = 0$: at rest
pump	G held from $t = 0$

Fully Developed Flow

Fully developed flow — the flow in a straight tube far from its bends, where the profile no longer changes along the tube. Keeping the volume then forces the velocity towards the wall to zero, the inertia term is exactly zero, and only $u(r,\ t)$ along the tube is left: Stokes is the full equation.

FROM NAVIER–STOKES TO THE PARABOLA, IN FIVE STEPS

$$\rho\frac{\partial \mathbf{u}}{\partial t}+\rho\,(\mathbf{u}\cdot\nabla)\mathbf{u}=-\nabla p+\mu\,\nabla^2\mathbf{u}$$

Navier–Stokes. Fully developed and $v=0$: the inertia term is zero

$$\rho\frac{\partial u}{\partial t}=-\frac{\partial p}{\partial x}+\mu\,\nabla^2 u=G+\mu\,\nabla^2 u$$

its x part. Straight tube: p falls evenly, $\partial p/\partial x = -G$

$$\nabla^2 u=\frac{\partial^2 u}{\partial x^2}+\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}$$

in tube coordinates: $\partial^2 u/\partial x^2 = 0$ (developed), $\partial^2 u/\partial \theta^2 = 0$ (round)

$$\rho\frac{\partial u}{\partial t}=G+\mu\,\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)$$

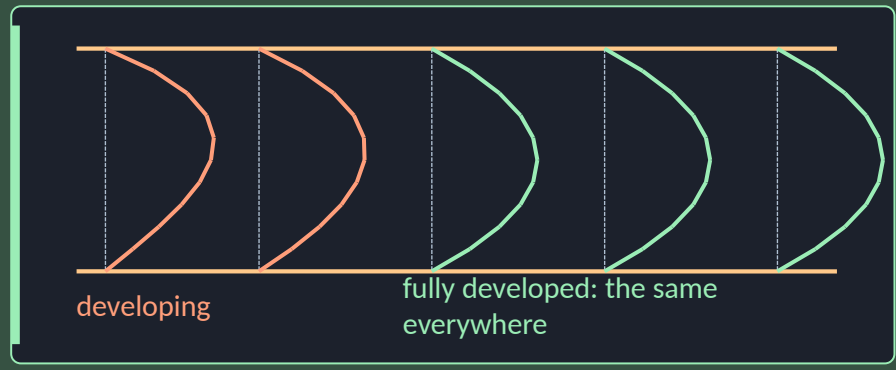
what is left: one unknown, $u(r,\ t)$: the start-up

$$0=G+\mu\,\frac{1}{r}\frac{d}{dr}\left(r\frac{du}{dr}\right)\ \Rightarrow\ u(r)=\frac{GR^2}{4\mu}\left(1-\frac{r^2}{R^2}\right)$$

steady, $\partial u/\partial t = 0$: integrated twice with the wall and axis conditions - the parabola

u the velocity along the tube, x along, r from the axis, θ round it, ∂ a partial derivative
G the pressure gradient, the drop per metre (Pa/m), μ the viscosity, ρ the density

DEVELOPING, THEN DEVELOPED



round, straight	the same at every θ: no swirl
far from bends	$\partial u/\partial x = 0$: fully developed
volume kept	then $v = 0$: only $u(r,\ t)$

Hagen–Poiseuille Flow

Hagen–Poiseuille flow — the steady flow in a fully developed tube: a parabola, fastest on the axis and zero on the wall. Its mean speed is Darcy's law with the permeability of a round tube, $K = R^2/8$, exact here because the inertia is zero; written as a friction factor, $f = 64/Re$.

THE PARABOLA, ITS MEAN, ITS FRICTION

$$u(r) = \frac{GR^2}{4\mu}\left(1 - \frac{r^2}{R^2}\right) \qquad f = \frac{64}{Re}$$

$$\bar{u} = \frac{GR^2}{8\mu} = \frac{K}{\mu}G, \quad K = \frac{R^2}{8}$$

R the tube's radius, G the pressure gradient (Pa/m), $\bar{u}$ the mean speed, K the permeability of a round tube
f Darcy's friction factor, Re the Reynolds number



$f = 64/Re$: friction falls as the flow speeds up

the cold plate: Re 452, f = 0.142

past 2300 the flow is turbulent: Ex_09.1 section 3

THE COLD PLATE'S TUBE, STEADY

quantity	value
mean speed $\bar{u}$	0.177 m/s
centre speed	0.354 m/s, twice the mean
pressure gradient G	393 Pa per metre
friction $f = 64/Re$	0.142
permeability K	1.1 mm²

Why a parabola: the pressure pushes every shell alike. The friction between shells carries the push of everything inside out to the wall, so the shear stress grows with r; $\tau = Gr/2$, and u falls from the axis to zero at the sticking wall. Without friction every parcel would go alike.

Start-Up Flow

Start-up flow — the same tube from rest, when the pump starts. Its exact solution is the parabola less a sum of terms that each die away at their own rate. Hagen–Poiseuille is not a second answer: it is where the start-up ends, after about R^2/ν = 3.8 s.

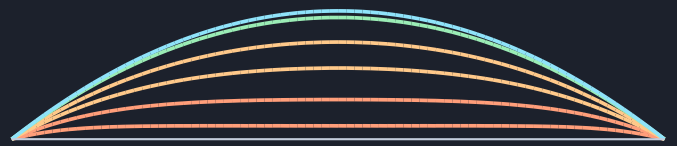
THE PARABOLA, LESS WHAT HAS NOT ARRIVED

$$u(r,t) = u(r) - \frac{GR^2}{4\mu}\sum_n c_n J_0\left(\frac{\lambda_n r}{R}\right) e^{-\lambda_n^2 \nu t/R^2}$$

$$c_n = \frac{8}{\lambda_n^3 J_1(\lambda_n)}, \quad J_0(\lambda_n) = 0$$

$$r^2 y'' + r y' + (r^2 - n^2) y = 0 \qquad \text{(Bessel's equation of order } n\text{)}$$

J_0, J_1 Bessel functions, λ_n the zeros of J_0 : 2.405, 5.520, 8.654, c_n the weight of each term, $\nu = \mu/\rho$
J_n is the solution of Bessel's equation finite at $r = 0$; here $n = 0$, from $u = R(r)\,T(t)$ in the tube equation

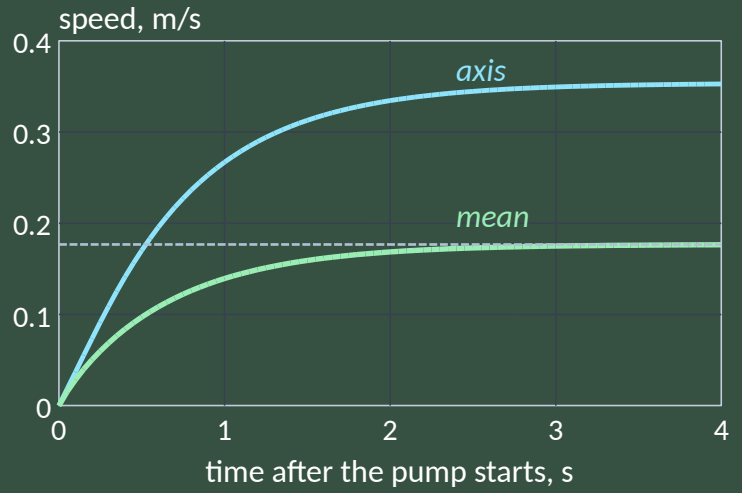


0.1 s: a flat core, pushed at once

1 s: rounding

4 s: the parabola

THE MEAN SPEED, AND THE AXIS



The mean speed - what a flow meter reads - is half-way after 0.43 s and within a per cent after about three. Ex_09.1's exact solution.

The Trial Function

Trial function — the network does not output the velocity; it outputs a free factor $\mathcal{N}$, and the velocity is built as $\hat{u} = (1 - s)\cdot\hat{t}\cdot\mathcal{N}$. The factor $1 - s$ is zero on the wall, $\hat{t}$ is zero at the start, so both conditions hold whatever $\mathcal{N}$ learns. The loss is then the equation alone.

WALL AND START BUILT IN: EX_09.1 TODO 1, 2

$$\hat{u}_{\hat{t}} = a + 4\kappa\,(s\,\hat{u}_{ss} + \hat{u}_s)$$

$$\hat{u} = (1 - s)\,\hat{t}\,\mathcal{N}(s, \hat{t})$$

$s = (r/R)^2$ where across the tube (0 axis, 1 wall), $\hat{t} = t/4\,\mathrm{s}$ when (0 pump start, 1 window end)
$\hat{u} = u/u_c$ the velocity over the steady centre speed, $\mathcal{N}$ the network, κ the viscosity and a the push, scaled

THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=2, n_hidden=32, n_layers=4)          # s and t in
u = (1 - s) * t * net(torch.cat([s, t], 1))        # wall and start, built in
us = grad(u, s); ut = grad(u, t)                   # derivatives by autograd
r = (ut - a) / kappa - 4 * (s * grad(us, s) + us)  # the residual, the whole loss
```

INPUTS, AND THE TWO FACTORS



$s = (r/R)^2$	where across the tube: 0 on the axis, 1 at the wall
$\hat{t} = t \,/\, 4\,\mathrm{s}$	time since the pump started, as a fraction of the 4 s start-up
$1 - s, \hat{t}$	zero on the wall and zero at the start: no slip and rest, built in

s is the same on both sides of the axis: the symmetry is in the input.

FDM or PINN

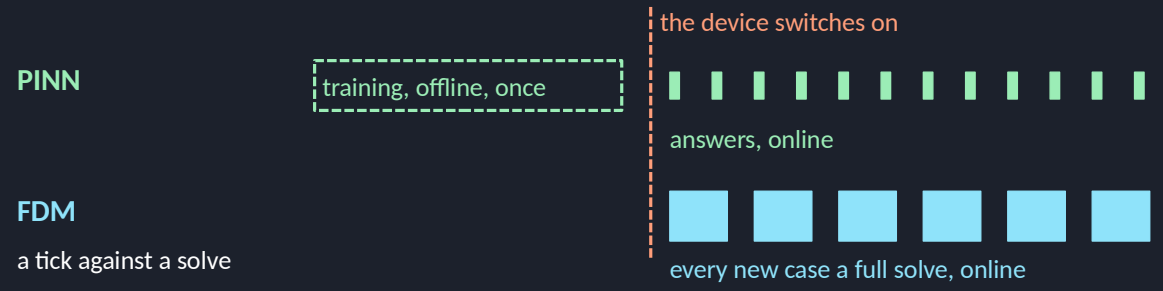
FDM or PINN — FDM computes an answer when asked. A PINN is trained before the device runs, offline; online, one evaluation is its answer. Compare online against online: one solve against one evaluation, same device, same accuracy, for a new case (a solved field can be stored).

ONLINE AGAINST ONLINE, AT THE SAME ACCURACY

$$S = \frac{t_{FDM}}{t_{PINN}} \quad \text{at the same error, on the same device}$$

t_{FDM} one finite-difference solve, t_{PINN} one evaluation of the trained network; the training is offline and not in S

OFFLINE ONCE, ONLINE EVERY TIME



EX_09.1 SECTION 5, ON COLAB'S T4

measured	FDM	PINN
online, one answer	one solve, 41 × 160	one evaluation
accuracy	1e-03, the agreed	scored the same way
device	the T4, synchronised	the T4, synchronised
offline	none	the training, apart
same case again	store the field, interpolate	no gain
new case	a full solve	one evaluation, if a parameter is an input

Run section 5 on Colab's T4 and read the numbers off its bar chart. On a problem this small the gap is tens, not thousands; the network's case is a new viscosity, push or speed per question, and the extra dimensions.

Viscosity From a Flow Meter

Inverse problem — the coolant's viscosity μ is not known - it changes threefold with temperature and glycol - but a flow meter's record is. Any model that predicts the readings can be fitted to them: the exact series, FDM inside an optimiser, or a PINN with μ trained beside its weights.

THE VISCOSITY IN THE LOSS, TRAINED WITH THE WEIGHTS

$$\mathcal{L}(\theta,\kappa)=\overline{\left[\frac{\hat{u}_{\hat{t}}-a}{\kappa}-4\,(s\,\hat{u}_{ss}+\hat{u}_{s})\right]^{2}}+w_d\sum_k\Big(\int_0^1\hat{u}\,ds-\bar{u}_k\Big)^2,\qquad \hat{u}=\hat{u}(s,\hat{t};\theta)$$

$$(\theta,\beta)\leftarrow(\theta,\beta)-\eta\,\nabla_{\theta,\beta}\mathcal{L},\qquad \kappa=e^{\beta},\qquad \mu=\kappa\frac{\rho R^2}{T}$$

θ the weights, inside $\hat{u}$; κ the scaled viscosity, inside the residual: $\mathcal{L}$ depends on both, so its gradient moves both $\beta=\ln\kappa$ keeps $\kappa>0$ (start 0, the nominal 2.5 mPa s), μ in Pa s ($T=4$ s), $\bar{u}_k$ the meter's reading at t_k, w_d its weight

THE INVERSE IN PYTORCH

```
beta = nn.Parameter(torch.tensor(0.0))  # beta = ln kappa; 0: nominal
kappa = beta.exp()                      # kappa > 0, always
loss = res(net, kappa).pow(2).mean() + w_d * misfit  # L(theta, kappa)
opt = Adam([*net.parameters(), beta])   # one gradient moves both
mu = kappa * rho * R**2 / T_END         # back to Pa s
```

FOUR WAYS, ONE RECORD

method (true μ 3.50)	μ [mPa·s]	time to the answer
Darcy on the last reading	4.04	none
the exact series, fitted	3.55	instant
FDM inside an optimiser	3.53	5 solves, 12 ms
the PINN, μ trained	3.56	46 s of training

Ex_09.1, the flow meter's first second. Darcy reads the last value as steady and is 15 % high, because the flow had not settled. Any model that predicts the readings can be fitted; the network is one of them, and it pays where no cheap solver exists.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
fronts, Reynolds ceiling	a thin feature against the points	Liu ch. 5.6	mixing layers, high Re
stream function	continuity built in exactly	Liu ch. 2.5.3	2-D flows past bodies
pressure reference	the level the equations leave free	Liu ch. 2.2	any closed duct
bends: Dean flow	the vortices of a curved tube	White ch. 6	serpentines, MP9.1A
porous media	Darcy flow through a solid	Liu ch. 2.4	groundwater, CO_2 storage
turbulent pipe flow	an averaged, modelled flow	White ch. 6	Re above 2300

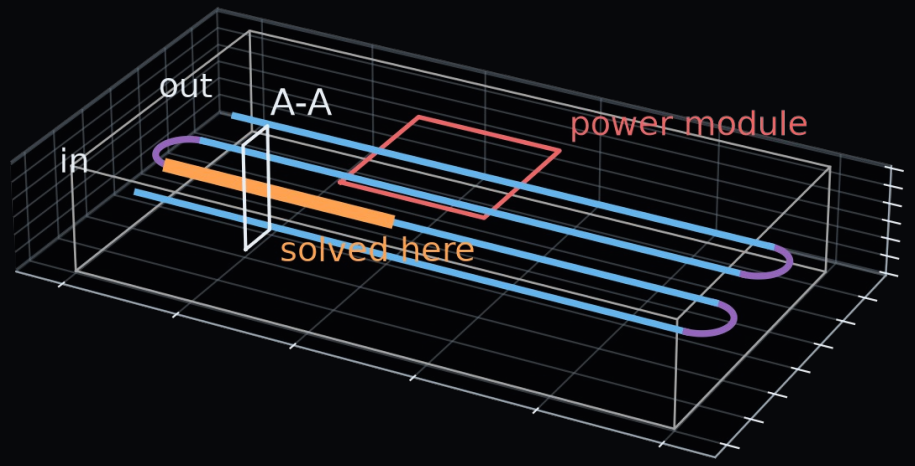
WHY THEY ARE LEFT OUT

Ex_09.1's tube is straight, laminar and fully developed, with an exact solution to score against, so none of them is needed to do it. The fourth row is mini project MP9.1A, where the students build the solver for a bend themselves.

Exercise 9.1

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the cold plate, the physics, the exact start-up
- **4** — the exact solution, and finite differences
- **5** — a PINN, conditions built in; timed at equal accuracy
- **6** — the coolant's viscosity from a flow meter
- **7 – 9** — the report; mini projects MP9.1A and MP9.1B



THE QUESTION EACH PART ANSWERS

1 – 3	what is left of Navier–Stokes?
4	which grid is fine enough?
5	when does a PINN pay?
6	which way finds the viscosity?

REPORT QUESTION, TO CONCLUDE

Section 5 timed one grid solve against one evaluation of the trained PINN, on the same device and at the same accuracy. Where in this cold plate would you choose the PINN, and what would you check first?

SUMMARY

Key Takeaways

PIPE FLOW	
what it is	Newton's law in a straight tube
function	the velocity across it, in time
train, tune	exact: parabola, Darcy, start-up
vs a bend	no inertia, no vortices
used for	pressure drop, pump, flow rate

SOLVING A FLOW	
what it is	a PINN, conditions built in
function	$u(s,\ t)$, and κ from data
train, tune	the equation alone; then L-BFGS
vs FDM	slower once, faster after N^*
used for	many queries, many dimensions

Next — L9.2: a gas in a fuel-cell channel, and what it carries.

Questions

1 – 2 · A NETWORK FOR THE TUBE

1 **How can a network learn the velocity of the coolant across the tube and in time after the pump starts?**

The network takes the position across the tube, as the squared radius, and the time since the pump started, and returns a factor from which the velocity is built. It is trained until the start-up equation of the straight tube holds at points spread over the section and the window; no solved data is needed. The wall, the start from rest and the symmetry about the axis are known exactly, so they are built into the form of the solution rather than learned. *Liu ch. 2.2; Ex_09.1 section 5*

2 **Why are the radius, the time and the velocity of the coolant scaled before the network is trained?**

In SI units the radius is three millimetres and the window four seconds, so the inputs differ by three orders of magnitude, and the second derivative across the tube is tens of thousands of times the velocity. A network of tanh neurons trains well on inputs and outputs of order one. Scaled by the radius, the window and the steady centre speed, every quantity is of order one, and two numbers carry the physics: the scaled viscosity and the push of the pressure gradient. *Ex_09.1 sections 3, 5*

3 – 4 · COST AND THE INVERSE

3 **When does a trained network give the velocity in the tube faster than finite differences, and how are the two compared?**

Online against online: one finite-difference solve against one evaluation of the trained network, on the same device and at the same accuracy, with the training reported apart as an offline cost. In this small tube the gain is tens, not thousands, and finite differences give a first answer far sooner because they need no training. The network pays when many answers are wanted for new cases, with a parameter such as the viscosity as an input, or in more dimensions. *Ex_09.1 section 5*

4 **How can the viscosity of the coolant be estimated with a network from a flow meter's readings, and is a network needed?**

The viscosity becomes a trainable number beside the weights, optimised with them so that the velocity satisfies the start-up equation and its mean matches the readings of the first second. The rise of the flow rate depends on the viscosity, which makes it identifiable. A network is not needed: the exact series or finite differences inside an optimiser fit the same readings faster here. Darcy's law on the last reading is 15 % high because the flow has not settled. *Liu ch. 2.2; Ex_09.1 section 6*

NEXT

L9.2 — Gas Flow

A gas in the fuel channel of a solid oxide cell: the flow is known, and the question is what it carries - hydrogen in, steam out - along the channel.

BEFORE THE NEXT SESSION

read	White ch. 6
run	Ex_09.1, sections 1 to 7
bring	your viscosity from the flow meter
think about	where you need many answers

A laminar flow, solved exactly and by two methods. Next, a flow that carries something along with it.